Different Components of Ground Tissue Systems are as follows

Hypodermis

One or two layers of continuous or discontinuous tissue present below the epidermis, is called hypodermis. It is protective in function.

In dicot stem, hypodermis is generally collenchymatous, whereas in monocot stem, it is generally sclerenchymatous. In many plants collenchyma form the hypodermis.

General Cortex

The Cortex occurs between the epidermis and pericycle. Cortex is a few to many layers in thickness, In most cases, it is made up of parenchymatous tissues. Intercellular spaces may or may not be present.

The cortical cells may contain non living inclusions of starch grains, oil, tannins and crystals.

Its general function is storage of food as well as providing mechanical support to organs.

Endodermis

The cells of this layer are barrel shaped and arranged compactly without intercellular spaces.

Endodermis is the innermost cortical layer that separates cortex from the stele.

Pericycle

Pericycle is single or few layered parenchymatous found inner to the endodermis. It is the outermost layer of the stele. Rarely thick walled sclerenchymatous. In angiosperms, pericycle gives rise to lateral roots.

Pith or Medulla

The central part of the ground tissue is known as pith or medulla. Generally this is made up of thin walled parenchyma cells with intercellular spaces. The cells in the pith generally stores starch, fatty substances, tannins, phenols, calcium oxalate crystals, etc.

Albuminous Cells: The cytoplasmic nucleated parenchyma, is associated with the sieve cells of Gymnosperms. Albuminous cells in *Conifers* are analogous to companion cells of Angiosperms. It also called as strasburger cells.

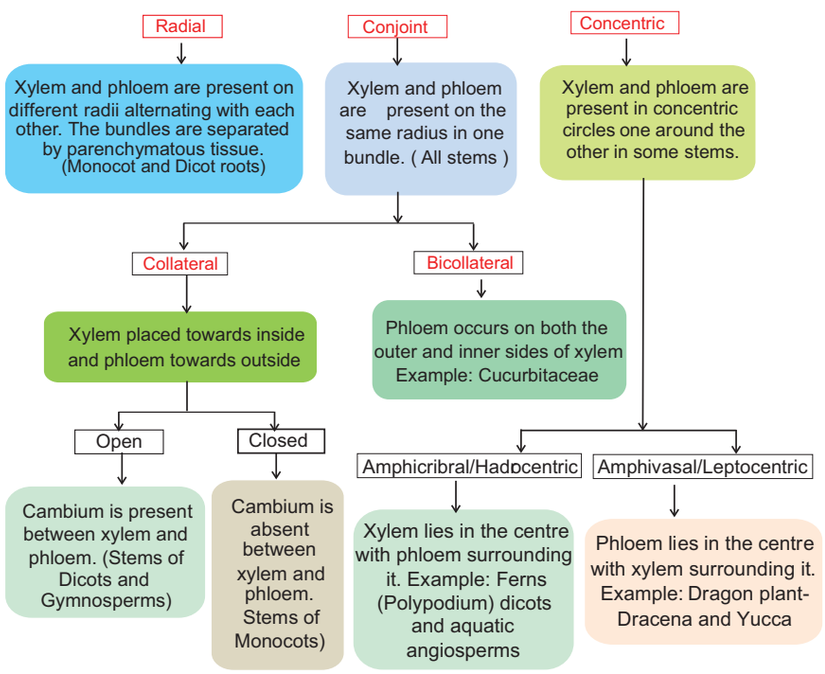
9.6 Vascular Tissue System

This section deals with the vascular tissue system of gymnosperms and angiosperms stems and roots.The vascular tissue system consists of xylem and phloem. The elements of xylem and phloem are always organized in groups. They are called **vascular bundles.**

The stems of both groups have an eustele while roots are protostele. In eustelic organization, the stele contains usually a ring of vascular bundles separated by interfascicular region or medullary ray

The structural and organizational variation in vascular bundles is shown below.

Types of vascular Bundles



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